Assignment 2

Tanu Kumari (251121)

1 Introduction

The objective of this project is to automatically detect and count buildings from satellite imagery using the YOLOv8 object detection framework. Such systems are useful for urban planning, disaster assessment, infrastructure monitoring, and mapping applications. The final pipeline accepts a satellite image as input and outputs detected buildings, their count, confidence scores, and an annotated image.

2 Dataset

The SpaceNet satellite imagery dataset was used for this project. The dataset contains high-resolution satellite images together with YOLO-format annotations representing building locations.

The dataset was divided into training, validation and testing subsets using an 80:10:10 split. The model was trained on a single object class:

- Building

During exploratory analysis, the dataset showed consistent image quality while presenting challenges such as dense building clusters, small rooftops and shadowed regions.

3 Model

YOLOv8s was selected because it provides an excellent balance between detection accuracy and inference speed. Instead of training from scratch, pretrained weights (yolov8s.pt) were used through transfer learning, enabling faster convergence and improved performance.

The YOLOv8s model contains approximately **11.13 million parameters**.

4 Training Configuration

Training was performed using Google Colab with an NVIDIA Tesla T4 GPU.

Parameter	Value
Model	YOLOv8s
Epochs	20
Batch Size	16
Image Size	640 × 640
GPU	Tesla T4
Final Train Box Loss	1.2591

Table 1: Training Configuration

5 Results

Metric	Value
Precision	0.819
Recall	0.725
F1 Score	0.769
mAP@50	0.804

Table 2: Evaluation Metrics

Figure 1 shows the variation of training and validation box loss over the training epochs. Both losses decrease steadily, indicating successful learning without severe overfitting.

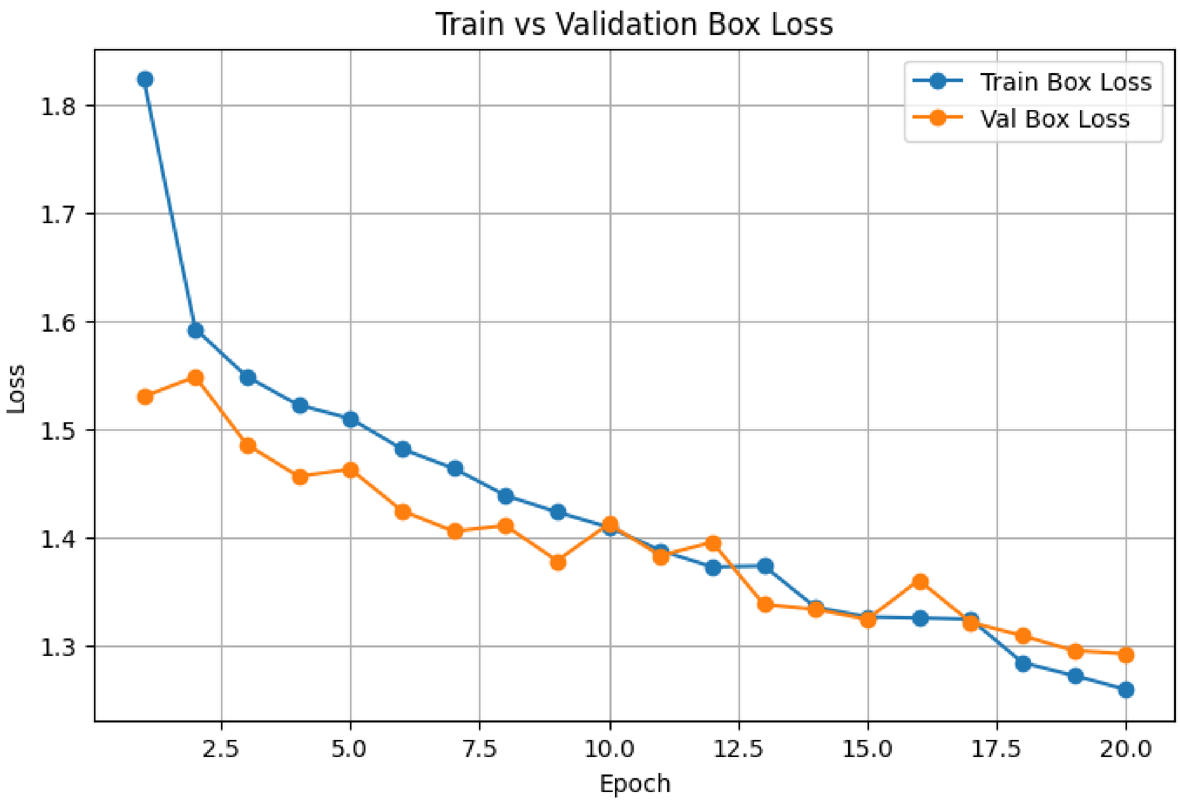

Figure 1: Training and Validation Box Loss

Figure 2 illustrates the improvement of mAP@50 during training. The metric increases with epochs and stabilizes towards the end of training.

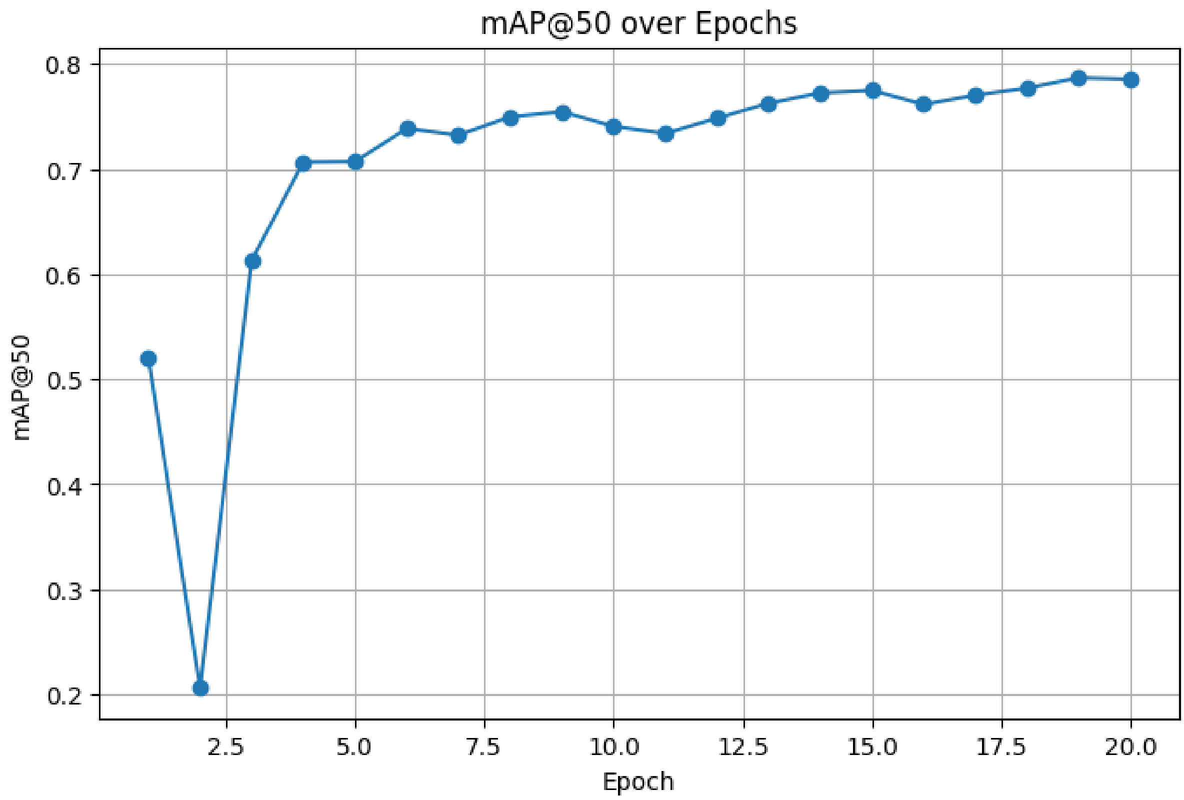

Figure 2: mAP@50 over Epochs

A sample prediction generated by the trained model is shown in Figure 3.

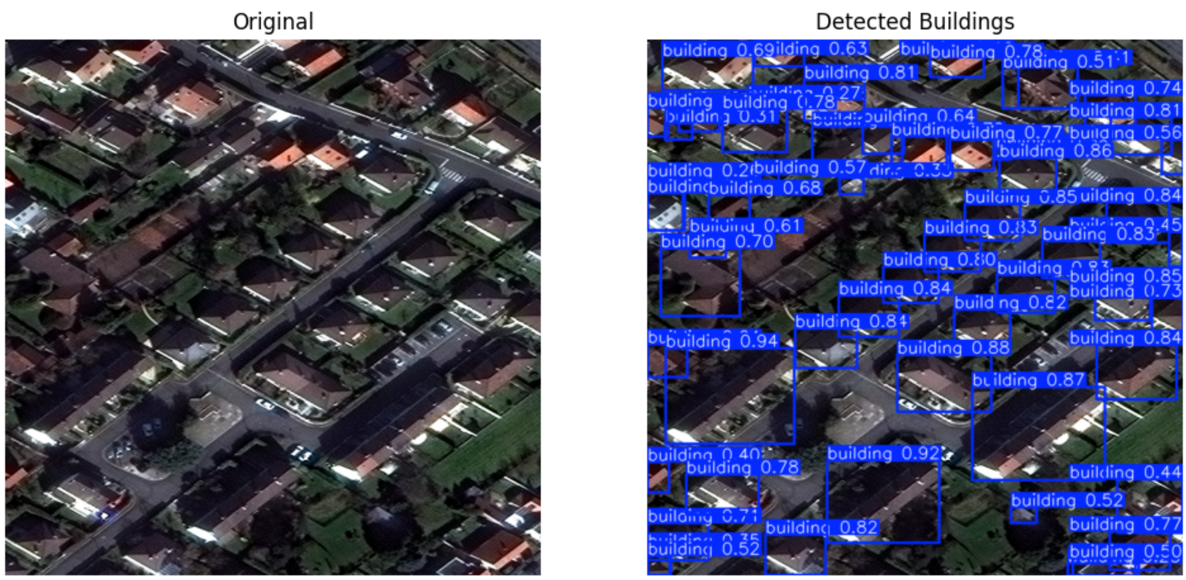

Figure 3: Sample Building Detection

6 Observations

The trained model achieved a Precision of 0.819 and an mAP@50 of 0.804, demonstrating good performance in detecting buildings from satellite imagery. The model performed well on images containing clearly separated buildings but faced challenges in dense urban regions, shadowed areas and images containing very small rooftops.

7 Conclusion

An end-to-end building detection pipeline was successfully developed using YOLOv8s. The model can automatically detect and count buildings in satellite imagery with good detection accuracy, providing a strong foundation for applications in urban mapping and geospatial analysis.